MILITARY SPECIFICATION

MODULES, STANDARD, ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE BMA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0760 failures per million operating hours for class I or 0.4771 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BMA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/253-1 for class I and M28787/253-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3 to 18 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

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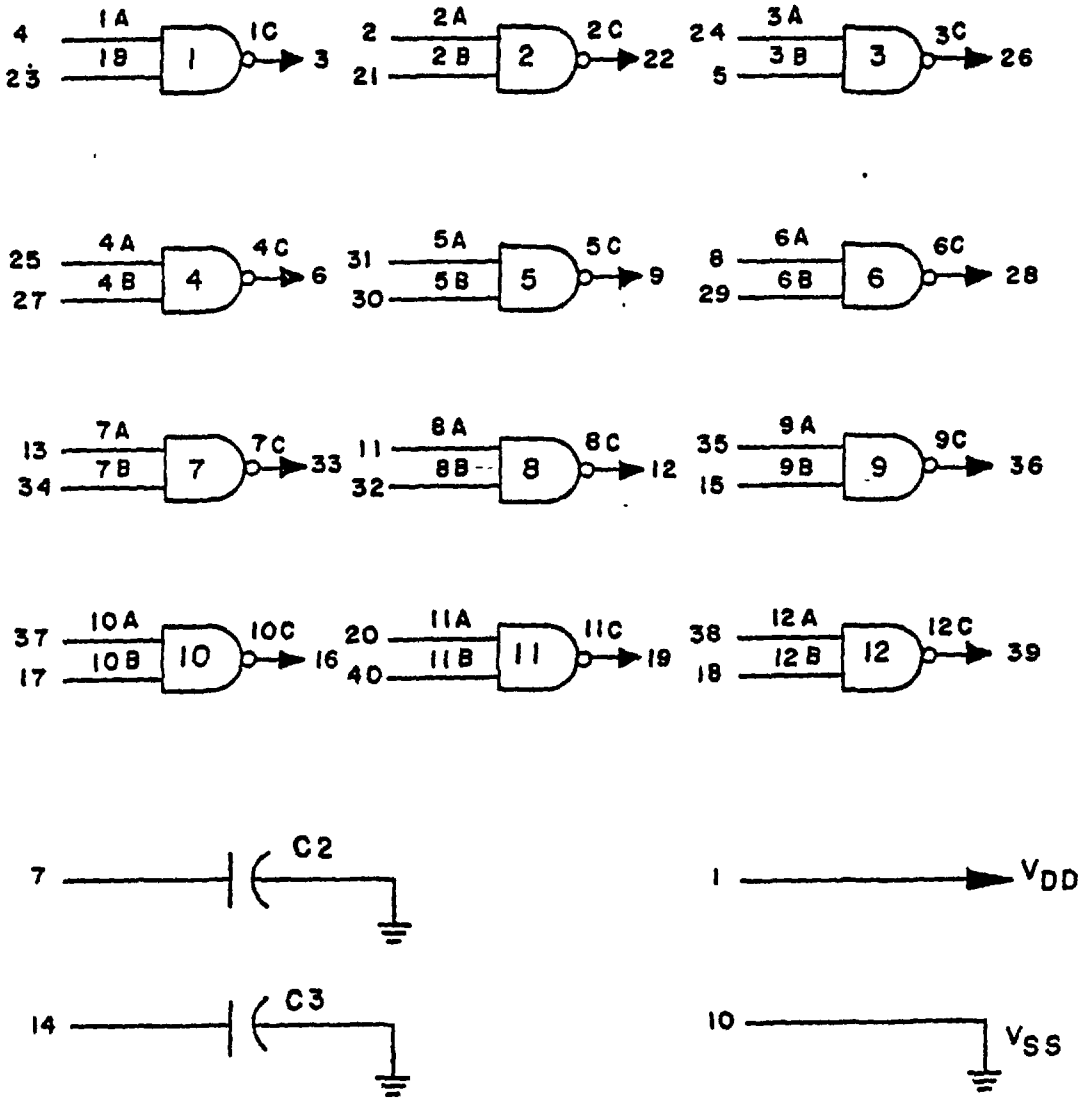


FIGURE 1. BMA module function diagram, NAND gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage (negative)	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage (positive)	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Output voltage (high level), I_{OH} = 0	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E12	Output voltage (low level), I_{OL} = 0	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F24	Output voltage (high level), V_{DD} = 5 V	VIII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
G1-G24	Output voltage (low level), V_{DD} = 5 V	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H12	Output voltage, (low level), V_{DD} = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J12	Output voltage (high level), V_{DD} = 15 V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
K1-K24	Input current (high level)	XII	X	0	200	0	200	0	200	0	200	0	200	nA
L1-L24	Input current (low level)	XIII	X	0	-200	0	-200	0	-200	0	-200	0	-200	nA
M1	Power supply current, low level	XIV	X	0	6	0	6	0	60	0	6	0	120	µA
M2	Power supply current, high level	XIV	X	0	6	0	6	0	60	0	6	0	120	µA
N1-N12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XV	S	20	200	20	200	22	220	15	150	24	245	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N13-N24	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XV	Q	68	680	68	680	75	750	51	515	83	835	ns
P1-P12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XVI	S	20	200	20	200	22	220	15	150	24	245	ns
P13-P24	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	68	680	68	680	75	750	51	515	83	835	ns
Q1-Q24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	25	250	25	250	27	275	19	190	30	305	ns
Q25-Q48	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	45	450	45	450	50	500	34	340	55	550	ns
P1-R24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XVIII	S	25	250	25	250	27	275	19	190	30	305	ns
R25-R48	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XVIII	Q	45	450	45	450	50	500	34	340	55	550	ns
S1-S12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	10	100	10	100	11	110	7	75	12	125	ns
T1-T12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XX	Q	10	100	10	100	11	110	7	75	12	125	ns
U1-U24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXI	Q	10	100	10	100	11	110	7	75	12	125	ns
V1-V24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXII	Q	10	100	10	100	11	110	7	75	12	125	ns
W1-W12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXIII	Q	8	80	8	80	9	90	6	60	10	100	ns
X1-X12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	8	80	8	80	9	90	6	60	10	100	ns

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXV	Q	8	80	8	80	9	90	6	60	10	100	ns
Z1-Z24	Propagation delay time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXVI	Q	8	80	8	80	9	90	6	60	10	100	ns
AA	Power supply filter capacitance	XXVII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1,BB2	C2 and C3 capacitance	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
CC1-CC24	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through J and M through CC shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions K and L shall be performed at 25°, 60° and 100°C only using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.1.2 Transient thermal test. For transient thermal test, use the same test limits as the maximum class temperature conditions.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

	Class I	Class II	Unit
Quiescent power dissipation	1.8 max	1.8 max	mW
Input voltage			
Low level V_{DD}= 5 V	0.0 min, 1.5 max	0.0 min, 1.5 max	V
V_{DD}= 15 V	0 min, 4 max	0 min, 4 max	V
High level V_{DD}= 5 V	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
V_{DD}= 15 V	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current (V_{DD} = 18 V)			
Low level (V_{IL} = 0 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5 V)			
I_{OL} (V_{OL} = 0.4V) at:			
-55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
I_{OH} (V_{OH} = 4.6V) at:			
-55°C	---	-0.64 min	mA
0° and 25°C	-0.51 min	-0.51 min	mA
60°C	-0.44 min	-0.44 min	mA
100°C	---	-0.36 min	mA

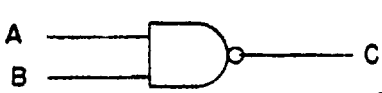
6.5 Module function. This module contains 12 independent 2 input NAND gates and 2 220 pF capacitors with 1 side connected to V_{SS} for external use.

6.5.1 Input to output relationship. Each gate shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Inputs		Outputs
A	B	C
L	L	H
L	H	H
H	L	H
H	H	L

NOTES:
1. H = High level.
2. L = Low level.



$C = \overline{A\ B}$

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6.5.2 Paralleling of outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. 1, 2, 3, and 4.

b. 5, 6, 7, and 8.

c. 9, 10, 11, and 12.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply V_{DD} is off.

d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.

e. Input voltages greater than the supply (V_{DD}-V_{SS}) should not be applied as this will forward bias the input protection diodes.

f. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

6.7 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{Total} = (CV_{DD}^2 f)N + \text{quiescent power dissipation.}$$

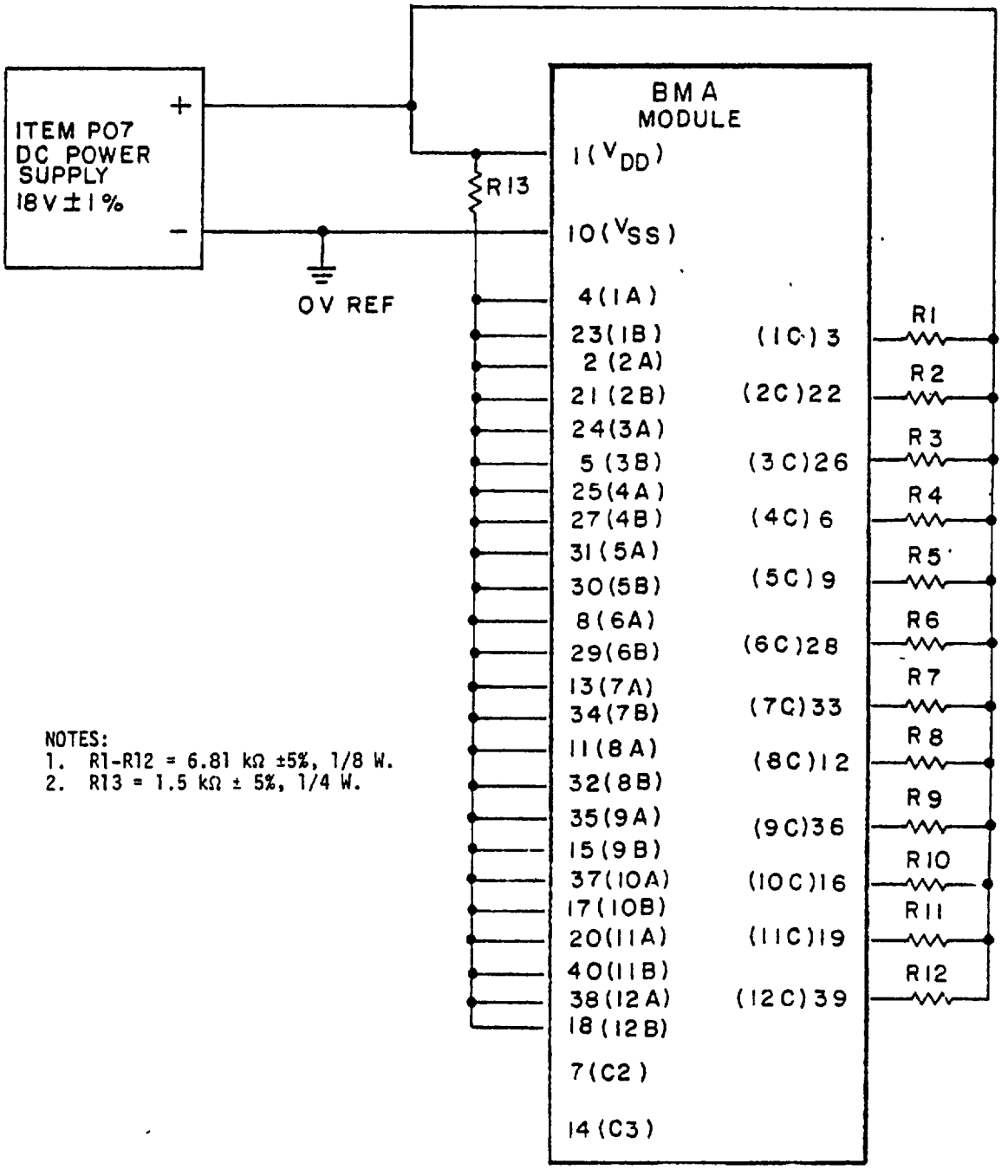
where C = Load capacitance.
V_{DD} = Supply voltage.
f = Operating frequency.
N = Number of outputs.

Preparing activity:
Navy - EC

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Agent:
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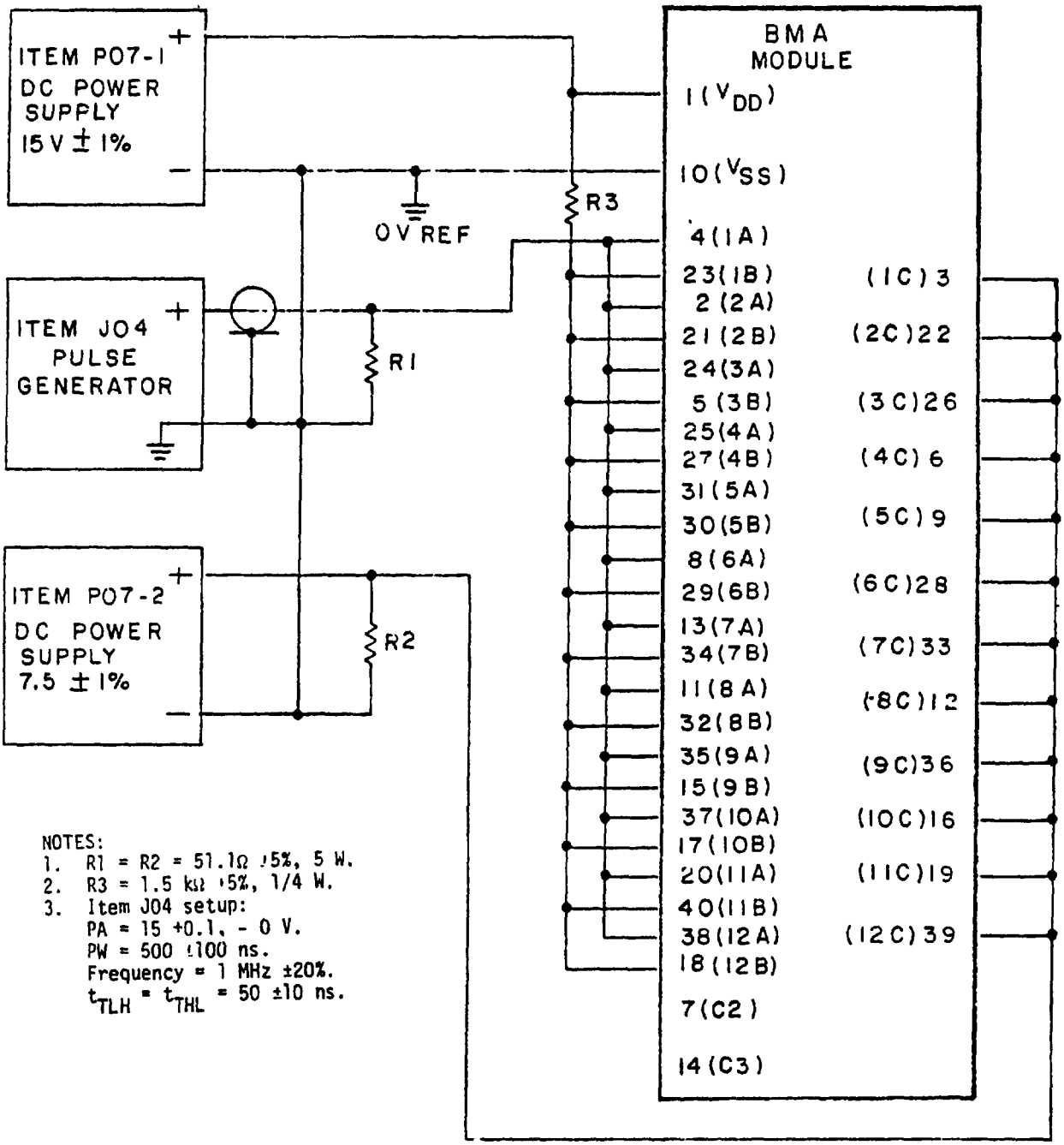
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NOTES:
1. R1-R12 = 6.81 kΩ ±5%, 1/8 W.
2. R13 = 1.5 kΩ ± 5%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 5 W.
2. R3 = 1.5 kΩ ±5%, 1/4 W.
3. Item J04 setup:
   PA = 15 +0.1, - 0 V.
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

FIGURE 3. Component temperature test circuit.

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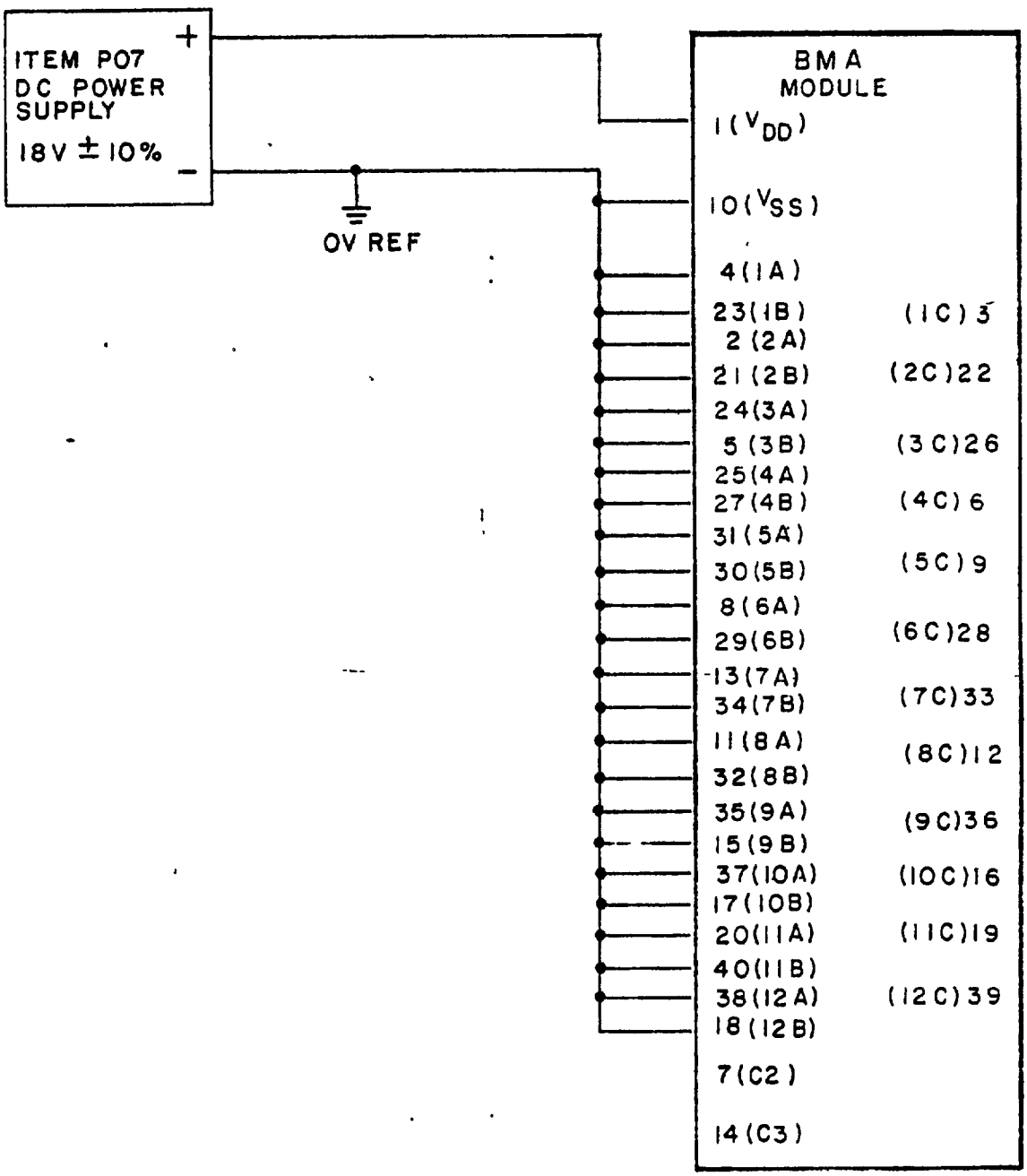


FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins	Frame	10	∞	MΩ	5

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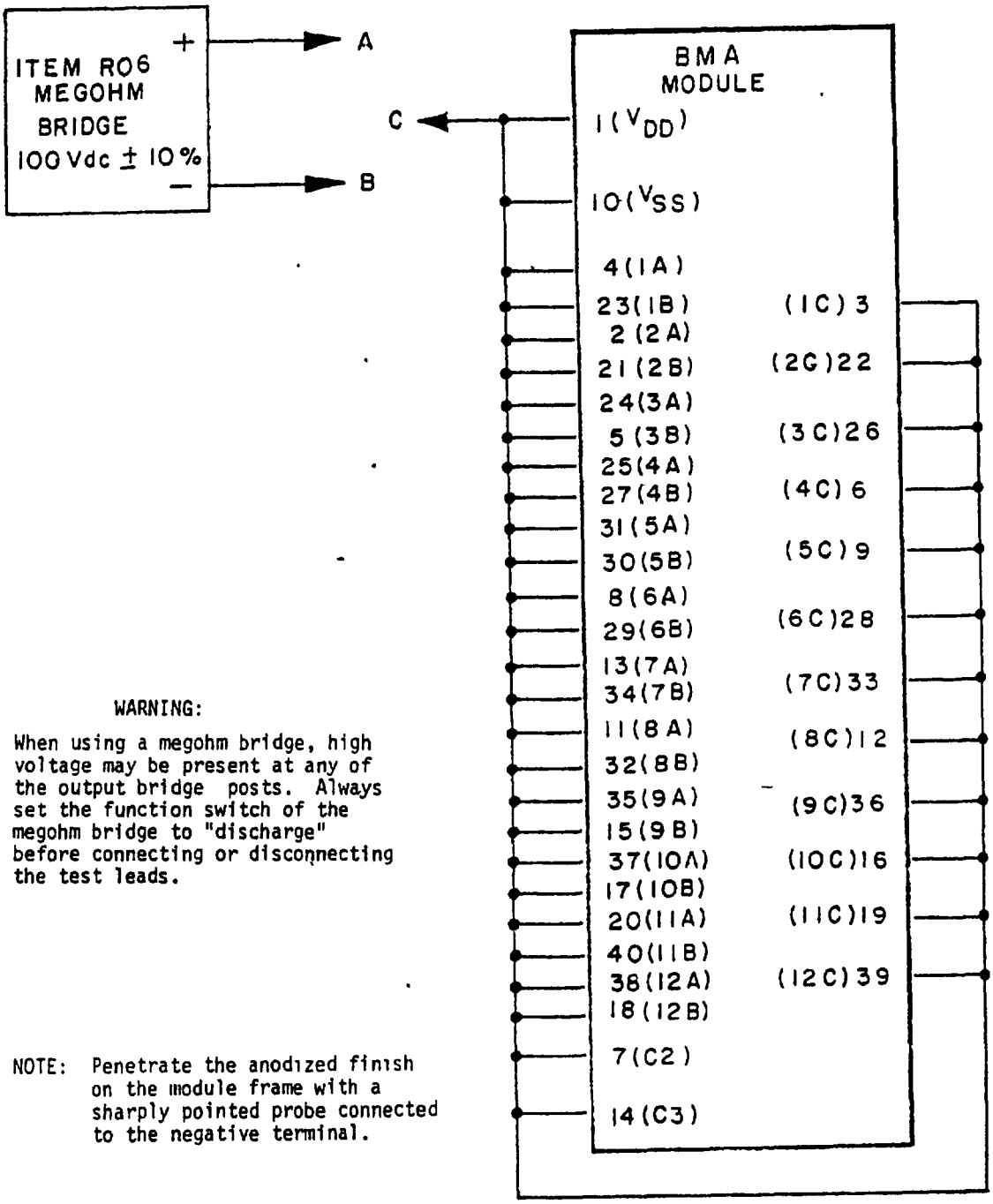


FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input clamp diode voltage (negative) test.

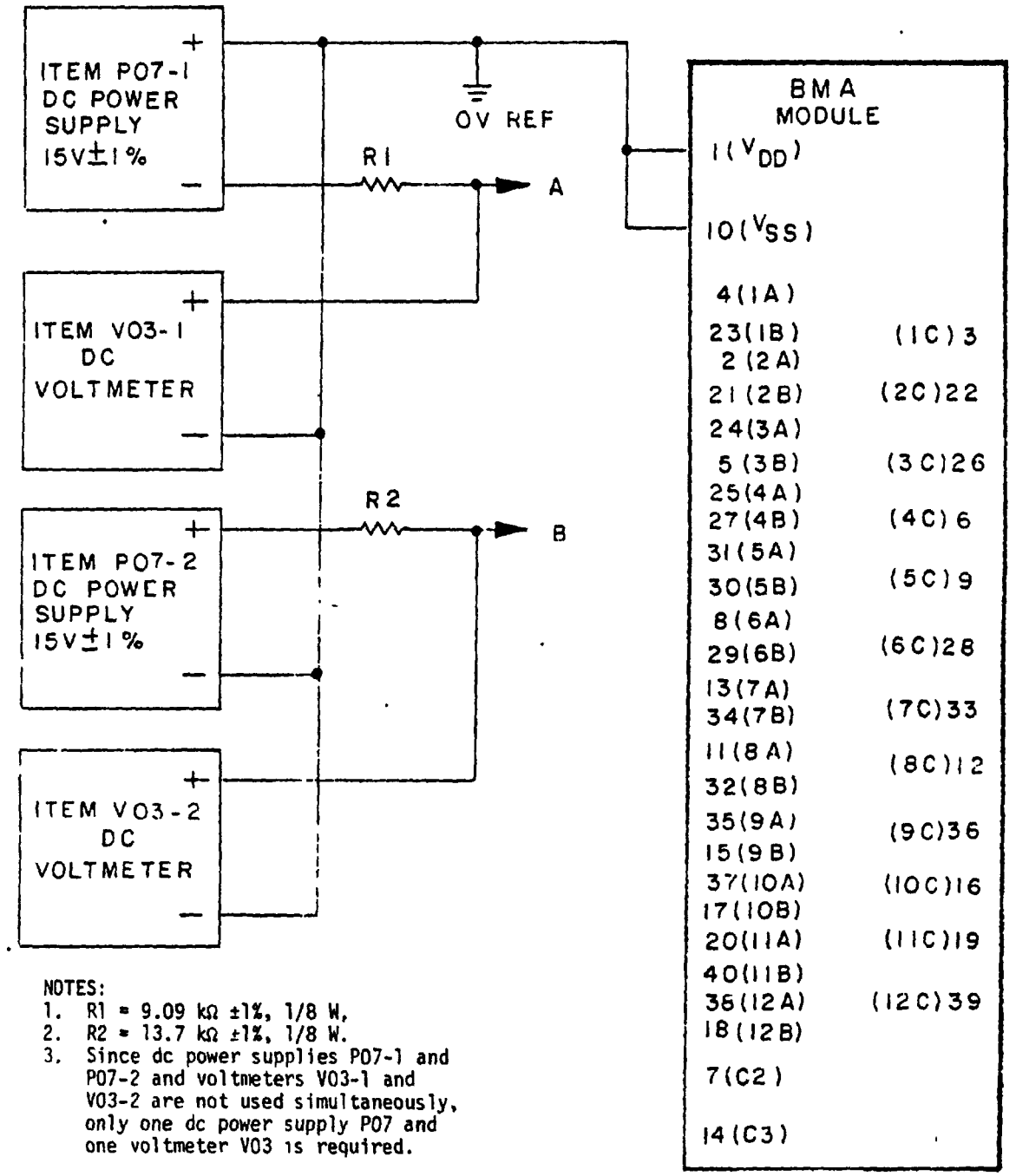
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input clamp diode voltage (negative)	4	Open	-0.4	-6.0	V	6
B2	1B	23					
B3	2A	2					
B4	2B	21					
B5	3A	24					
B6	3B	5					
B7	4A	25					
B8	4B	27					
B9	5A	31					
B10	5B	30					
B11	6A	8					
B12	6B	29					
B13	7A	13					
B14	7B	34					
B15	8A	11					
B16	8B	32					
B17	9A	35					
B18	9B	15					
B19	10A	37					
B20	10B	17					
B21	11A	20					
B22	11B	40					
B23	12A	38					
B24	12B	18					

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TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage (positive)	Open	4	0.4	1.5	V	6
C2	1B		23				
C3	2A		2				
C4	2B		21				
C5	3A		24				
C6	3B		5				
C7	4A		25				
C8	4B		27				
C9	5A		31				
C10	5B		30				
C11	6A		8				
C12	6B		29				
C13	7A		13				
C14	7B		34				
C15	8A		11				
C16	8B		32				
C17	9A		35				
C18	9B		15				
C19	10A		37				
C20	10B		17				
C21	11A		20				
C22	11B		40				
C23	12A		38				
C24	12B		18				

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W,
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since dc power supplies P07-1 and P07-2 and voltmeters V03-1 and V03-2 are not used simultaneously, only one dc power supply P07 and one voltmeter V03 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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TABLE VI. Output voltage (high-level, $I_{OH} = 0$) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1C Output voltage (high-level), $I_{OH} = 0$	3	10	4.95	5.05	V	7
D2	2C	22					
D3	3C	26					
D4	4C	6					
D5	5C	9					
D6	6C	28					
D7	7C	33					
D8	8C	12					
D9	9C	36					
D10	10C	16					
D11	11C	19					
D12	12C	39					

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TABLE VII. Output voltage (low-level, I_{OL} = 0) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1C Output voltage (low-level), I_{OL} = 0	3	1	0.00	0.05	V	7
E2	2C	22					
E3	3C	26					
E4	4C	6					
E5	5C	9					
E6	6C	28					
E7	7C	33					
E8	8C	12					
E9	9C	36					
E10	10C	16					
E11	11C	19					
E12	12C	39					

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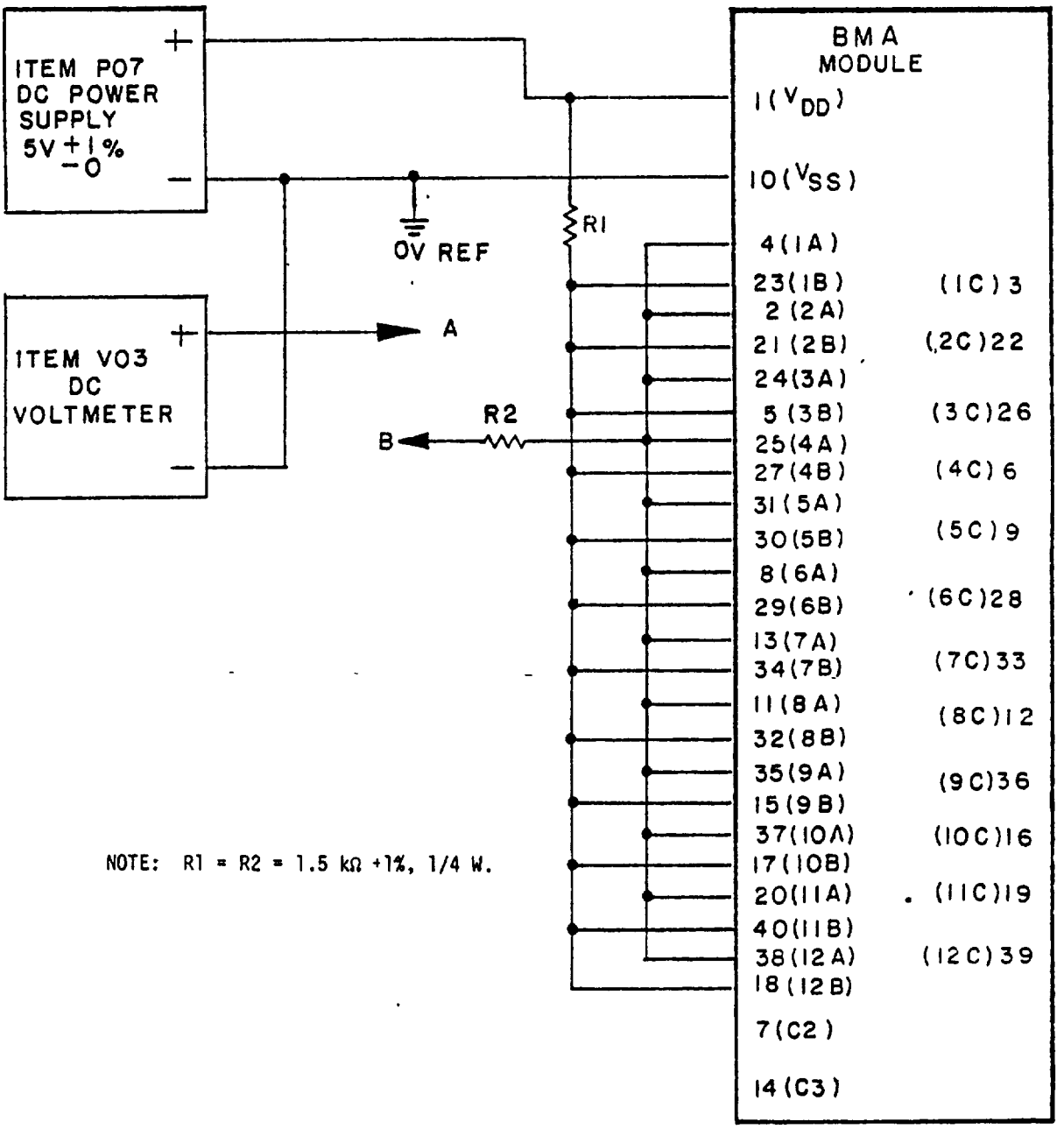


FIGURE 7. Output voltage (high-level, I_{OH} = 0 and low-level, I_{OL} = 0) test circuit for tables VI and VII.

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TABLE VIII. Output voltage (high-level, V_{DD} = 5 V) test.

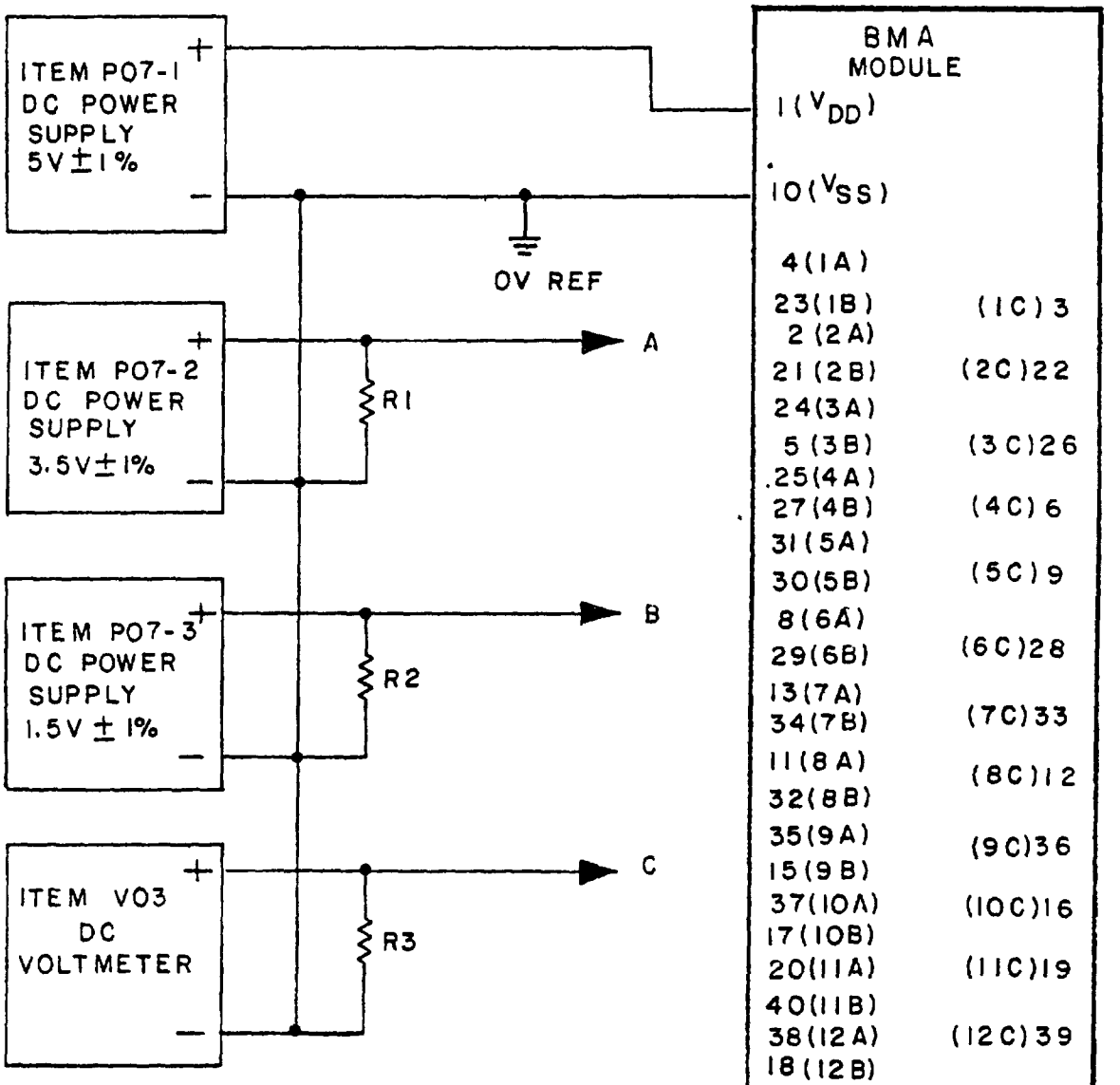
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1C Output voltage (high level), V_{DD} = 5 V	2,21,24,5,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18,4	23	3	4.6	5.0	V	8
F2	2C	4,23,24,5,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18,2	21	22				
F3	3C	4,23,2,21,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18,24	5	26				
F4	4C	4,23,2,21,24,5,31,30,8,29, 13,34,11,32,35,15,37,17, 20,40,38,18,25	27	6				
F5	5C	4,23,2,21,24,5,25,27,8,29, 13,34,11,32,35,15,37,17,20, 40,38,18,31	30	9				
F6	6C	4,23,2,21,24,5,25,27,31,30, 13,34,11,32,35,15,37,17,20, 40,38,18,8	29	28				
F7	7C	4,23,2,21,24,5,25,27,31,30, 8,29,11,32,35,15,37,17,20, 40,38,18,13	34	33				
F8	8C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,35,15,37,17,20, 40,38,18,11	32	12				
F9	9C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,37,17,20, 40,38,18,35	15	36				
F10	10C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,35,15,20, 40,38,18,37	17	16				
F11	11C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,35,15,37, 17,38,18,20	40	19				
F12	12C	4,23,2,21,24,5,25,27,31,30, 8,29,13,34,11,32,35,15,37, 17,20,40,38	18	39				

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TABLE VIII. Output voltage (high-level, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F13	1C Output voltage (high level), V_{DD} = 5 V	2,21,24,5,25,27,31,30,8,29,13,34,11,32,35,15,37,17,20,40,38,18,23	4	3	4.6	5.0	V	8
F14	2C	4,23,24,5,25,27,31,30,8,29,13,34,11,32,35,15,37,17,20,40,38,18,21	2	22				
F15	3C	4,23,2,21,25,27,31,30,8,29,13,34,11,32,35,15,37,17,20,40,38,18,5	24	26				
F16	4C	4,23,2,21,24,5,31,30,8,29,13,34,11,32,35,15,37,17,20,40,38,18,27	25	6				
F17	5C	4,23,2,21,24,5,25,27,8,29,13,34,11,32,35,15,37,17,20,40,38,18,30	31	9				
F18	6C	4,23,2,21,24,5,25,27,31,30,13,34,11,32,35,15,37,17,20,40,38,18,29	8	28				
F19	7C	4,23,2,21,24,5,25,27,31,30,8,29,11,32,35,15,37,17,20,40,38,18,34	13	33				
F20	8C	4,23,2,21,24,5,25,27,31,30,8,29,13,34,35,15,37,17,20,40,38,18,32	11	12				
F21	9C	4,23,2,21,24,5,25,27,31,30,8,29,13,34,11,32,37,17,20,40,38,18,15	35	36				
F22	10C	4,23,2,21,24,5,25,27,31,30,8,29,13,34,11,32,35,15,20,40,38,18,17	37	16				
F23	11C	4,23,2,21,24,5,25,27,31,30,8,29,13,34,11,32,35,15,37,17,38,18,40	20	19				
F24	12C	4,23,2,21,24,5,25,27,31,30,8,29,13,34,11,32,35,15,37,17,20,40,18	38	39				

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = At 0° and 25°C - 23.2 kΩ ±1%, 1/8 W.
   = At 60°C - 27.4 kΩ ±1%, 1/8 W.
   = At -55°C - 18.2 kΩ ±1%, 1/8 W.
   = At 100°C - 32.4 kΩ ±1%, 1/8 W.

FIGURE 8. Output voltage (high-level, V_{DD} = 5 V) test circuit for table VIII.

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TABLE IX. Output voltage (low-level, V_{DD} = 5 V) test.

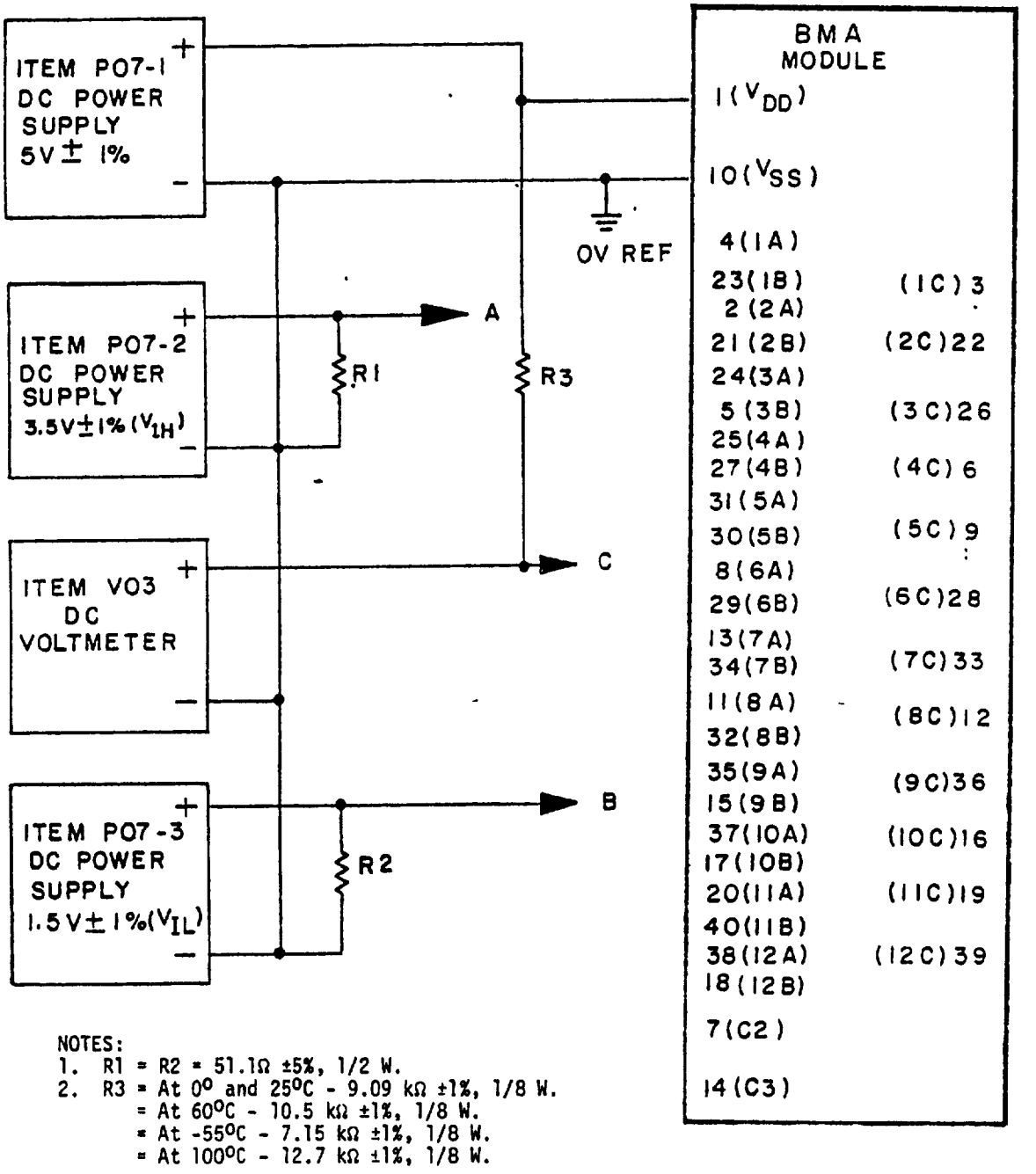
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1C Output voltage (low level), V_{DD} = 5 V	4	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	3	0.0	0.4	V	9
G2	1C	23	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	3				
G3	2C	2	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	22				
G4	2C	21	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	22				
G5	3C	24	4,23,2,21,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	26				
G6	3C	5	4,23,2,21,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	26				
G7	4C	25	4,23,2,21,24,5,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	6				
G8	4C	27	4,23,2,21,24,5,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	6				
G9	5C	31	4,23,2,21,24,5,25,27,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	9				
G10	5C	30	4,23,2,21,24,5,25,27,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	9				
G11	6C	8	4,23,2,21,24,5,25,27,31,30,29,13, 34,11,32,35,15,37,17,20,40,38,18	28				
G12	6C	29	4,23,2,21,24,5,25,27,31,30,8,13, 34,11,32,35,15,37,17,20,40,38,18	28				
G13	7C	13	4,23,2,21,24,5,25,27,31,30,8,29, 34,11,32,35,15,37,17,20,40,38,18	33				
G14	7C	34	4,23,2,21,24,5,25,27,31,30,8,29, 13,11,32,35,15,37,17,20,40,38,18	33				
G15	8C	11	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,32,35,15,37,17,20,40,38,18	12				
G16	8C	32	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,35,15,37,17,20,40,38,18	12				

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TABLE IX. Output voltage (low-level, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G17	9C Output voltage (low level), V_{DD} = 5 V	35	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,15,37,17,20,40,36,18	36	0.0	0.4	V	9
G18	9C	15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,37,17,20,40,38,18	36				
G19	10C	37	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,17,20,40,38,18	16				
G20	10C	17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,20,40,38,18	16				
G21	11C	20	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,40,38,18	19				
G22	11C	40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,38,18	19				
G23	12C	38	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,18	39				
G24	12C	18	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38	39				

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = At 0° and 25°C - 9.09 kΩ ±1%, 1/8 W.
   = At 60°C - 10.5 kΩ ±1%, 1/8 W.
   = At -55°C - 7.15 kΩ ±1%, 1/8 W.
   = At 100°C - 12.7 kΩ ±1%, 1/8 W.

FIGURE 9. Output voltage (low-level, V_{DD} = 5 V) test circuit for table IX.

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TABLE X. Output voltage (low-level, V_{DD} = 15 V) test.

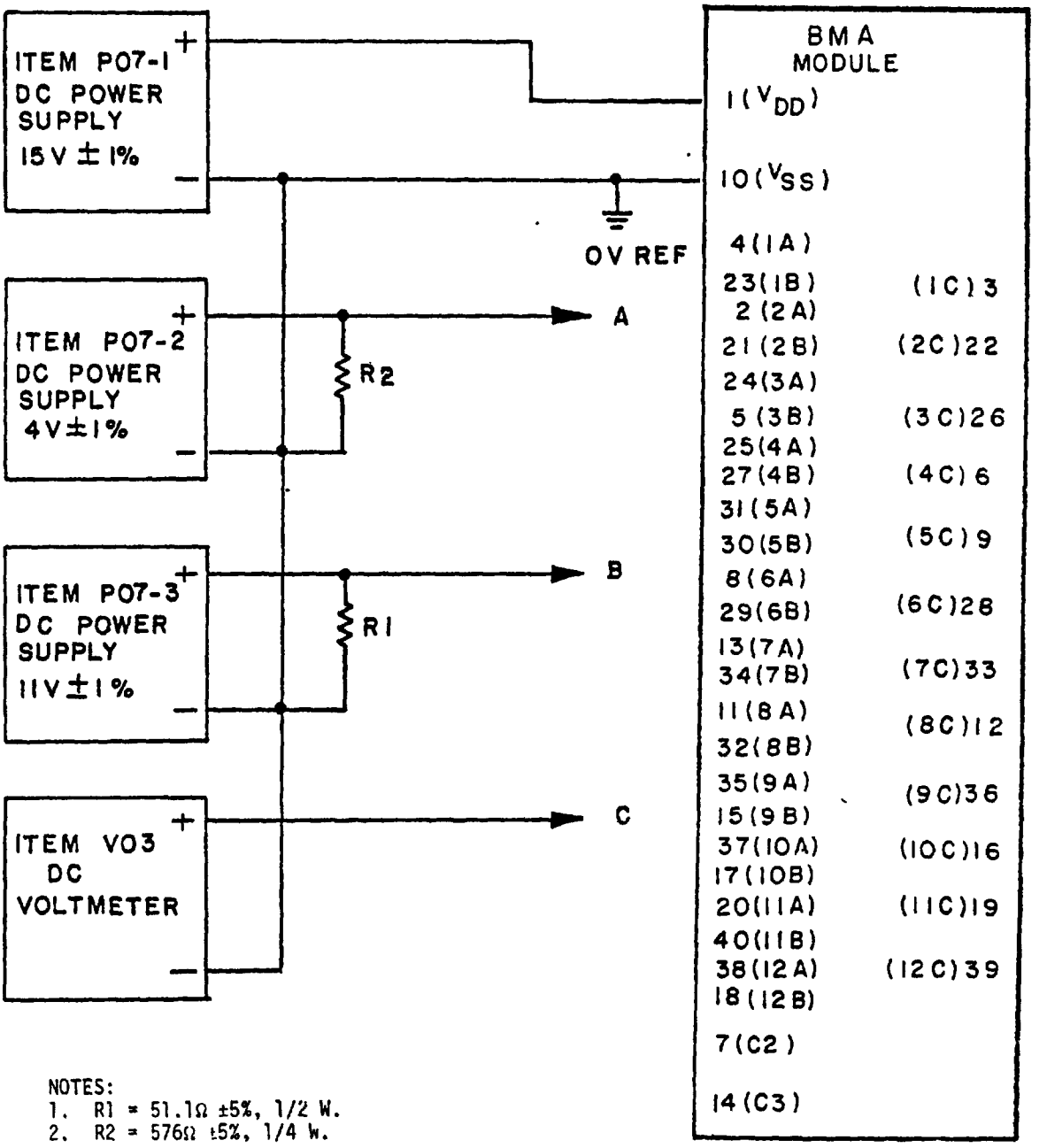
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1C Output voltage (low level), V_{DD} = 15 V	Open	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,32,35,15,37,17,20,40, 38,18	3	0.0	1.5	V	10
H2	2C			22				
H3	3C			26				
H4	4C			6				
H5	5C			9				
H6	6C			28				
H7	7C			33				
H8	8C			12				
H9	9C			36				
H10	10C			16				
H11	11C			19				
H12	12C			39				

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TABLE XI. Output voltage (high-level, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1C Output voltage (high level), V_{DD} = 15 V	23,21,5,27,30,29, 34,32,15,17,40,18	4,2,24,25,31,8,13,11, 35,37,20,38	3	13.5	15.0	V	1C
J2	2C			22				
J3	3C			26				
J4	4C			6				
J5	5C			9				
J6	6C			28				
J7	7C			33				
J8	8C			12				
J9	9C			36				
J10	10C			16				
J11	11C			19				
J12	12C			39				

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/4 W.

FIGURE 10. Output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables X and XI.

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TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1A Input current (high level)	4	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	0	200	nA	11
K2	1B	23	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K3	2A	2	4,23,21,24,5,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
K4	2B	21	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K5	3A	24	4,23,2,21,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K6	3B	5	4,23,2,21,24,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
K7	4A	25	4,23,2,21,24,5,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K8	4B	27	4,23,2,21,24,5,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K9	5A	31	4,23,2,21,24,5,25,27,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K10	5B	30	4,23,2,21,24,5,25,27,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
K11	6A	8	4,23,2,21,24,5,25,27,31,30,29,13 34,11,32,35,15,37,17,20,40,38,18				
K12	6B	29	4,23,2,21,24,5,25,27,31,30,8,13, 34,11,32,35,15,37,17,20,40,38,18				
K13	7A	13	4,23,2,21,24,5,25,27,31,30,8,29, 34,11,32,35,15,37,17,20,40,38,18				
K14	7B	34	4,23,2,21,24,5,25,27,31,30,8,29, 13,11,32,35,15,37,17,20,40,38,18				
K15	8A	11	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,32,35,15,37,17,20,40,38,18				
K16	8B	32	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,35,15,37,17,20,40,38,18				

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K17	9A Input current (high level)	35	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,15,37,17,20,40,38,18	0	200	nA	11
K18	9B	15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,37,17,20,40,38,18				
K19	10A	37	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,17,20,40,38,18				
K20	10B	17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,20,40,38,18				
K21	11A	20	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,40,38,18				
K22	11B	40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,38,18				
K23	12A	38	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,18				
K24	12B	18	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38				

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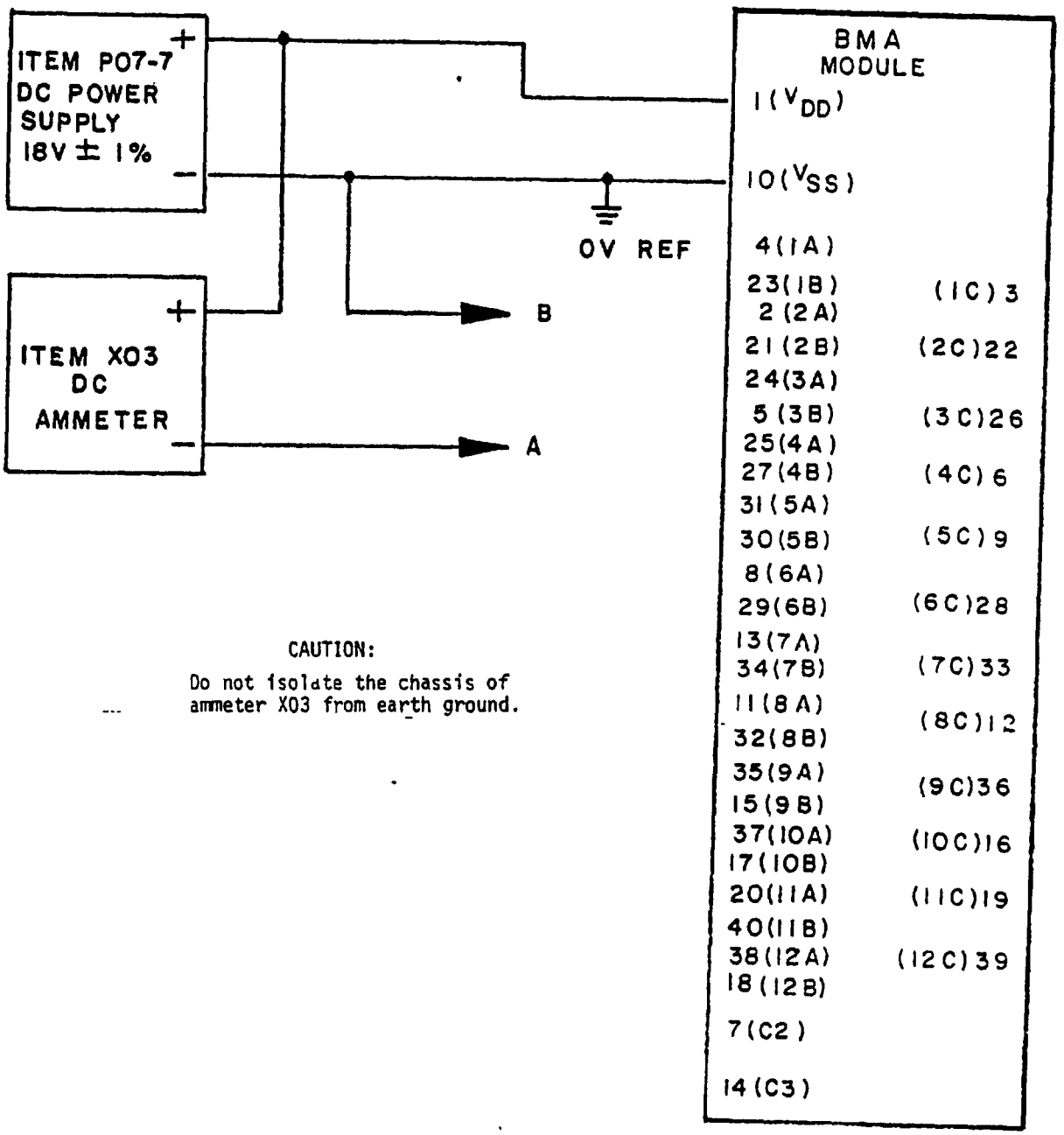


FIGURE 11. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

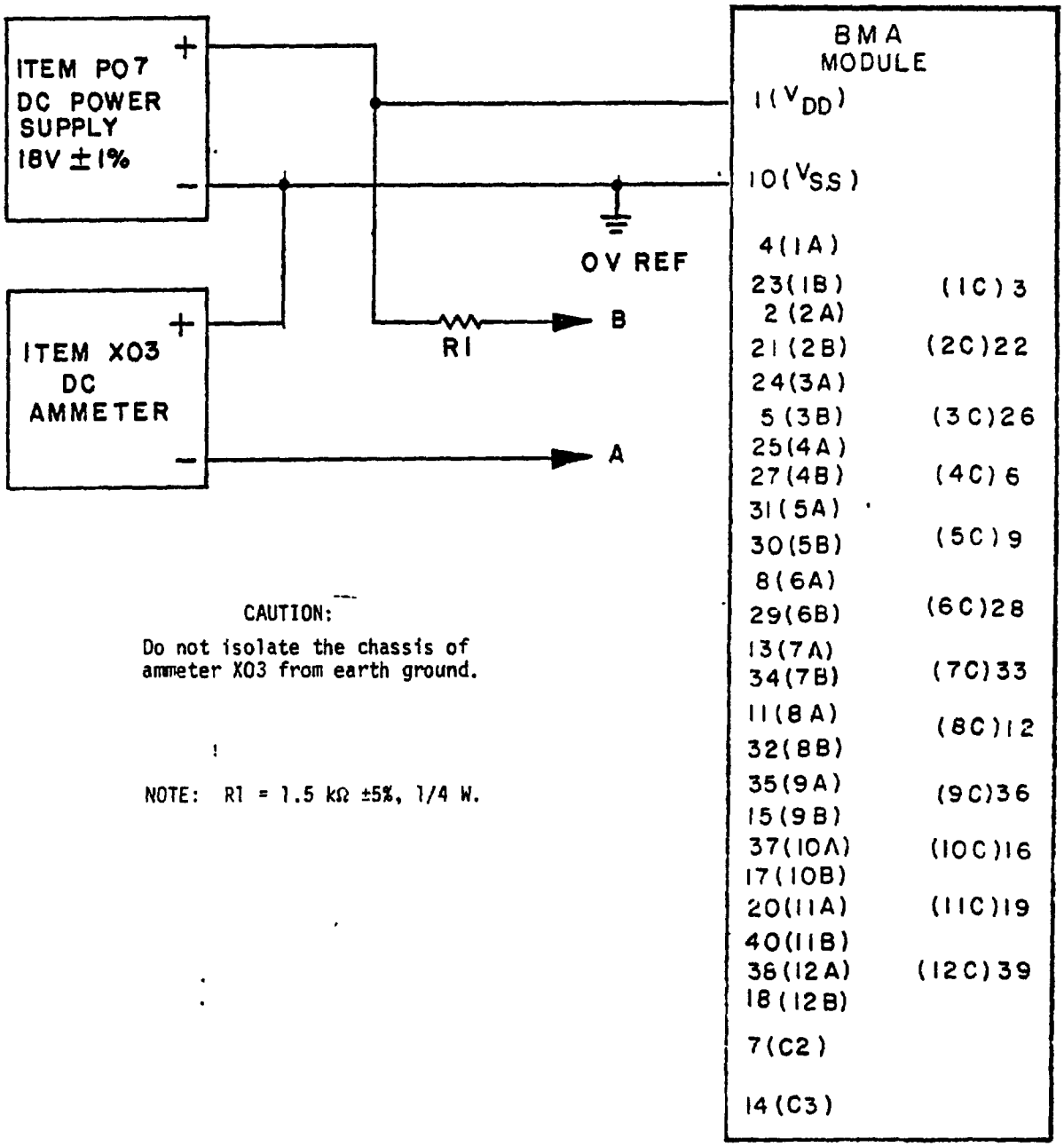
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1A Input current (low level)	4	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	0.0	-200	nA	12
L2	1B	23	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L3	2A	2	4,23,21,24,5,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
L4	2B	21	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L5	3A	24	4,23,2,21,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L6	3B	5	4,23,2,21,24,25,27,31,30,8,29,13 34,11,32,35,15,37,17,20,40,38,18				
L7	4A	25	4,23,2,21,24,5,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L8	4B	27	4,23,2,21,24,5,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L9	5A	31	4,23,2,21,24,5,25,27,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L10	5B	30	4,23,2,21,24,5,25,27,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18				
L11	6A	8	4,23,2,21,24,5,25,27,31,30,29,13 34,11,32,35,15,37,17,20,40,38,18				
L12	6B	29	4,23,2,21,24,5,25,27,31,30,8,13, 34,11,32,35,15,37,17,20,40,38,18				
L13	7A	13	4,23,2,21,24,5,25,27,31,30,8,29, 34,11,32,35,15,37,17,20,40,38,18				
L14	7B	34	4,23,2,21,24,5,25,27,31,30,8,29, 13,11,32,35,15,37,17,20,40,38,18				
L15	8A	11	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,32,35,15,37,17,20,40,38,18				
L16	8B	32	4,23,2,21,24,5,25,27,31,30,8,29, 13,34,11,35,15,37,17,20,40,38,18				

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TABLE XIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L17	9A Input current (low level)	35	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,15,37,17,20,40,38,18	0.0	-200	nA	12
L18	9B	15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,37,17,20,40,38,18				
L19	10A	37	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,17,20,40,38,18				
L20	10B	17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,20,40,38,18				
L21	11A	20	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,40,38,18				
L22	11B	40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,38,18				
L23	12A	38	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,18				
L24	12B	16	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38				

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CAUTION:

Do not isolate the chassis of ammeter XO3 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

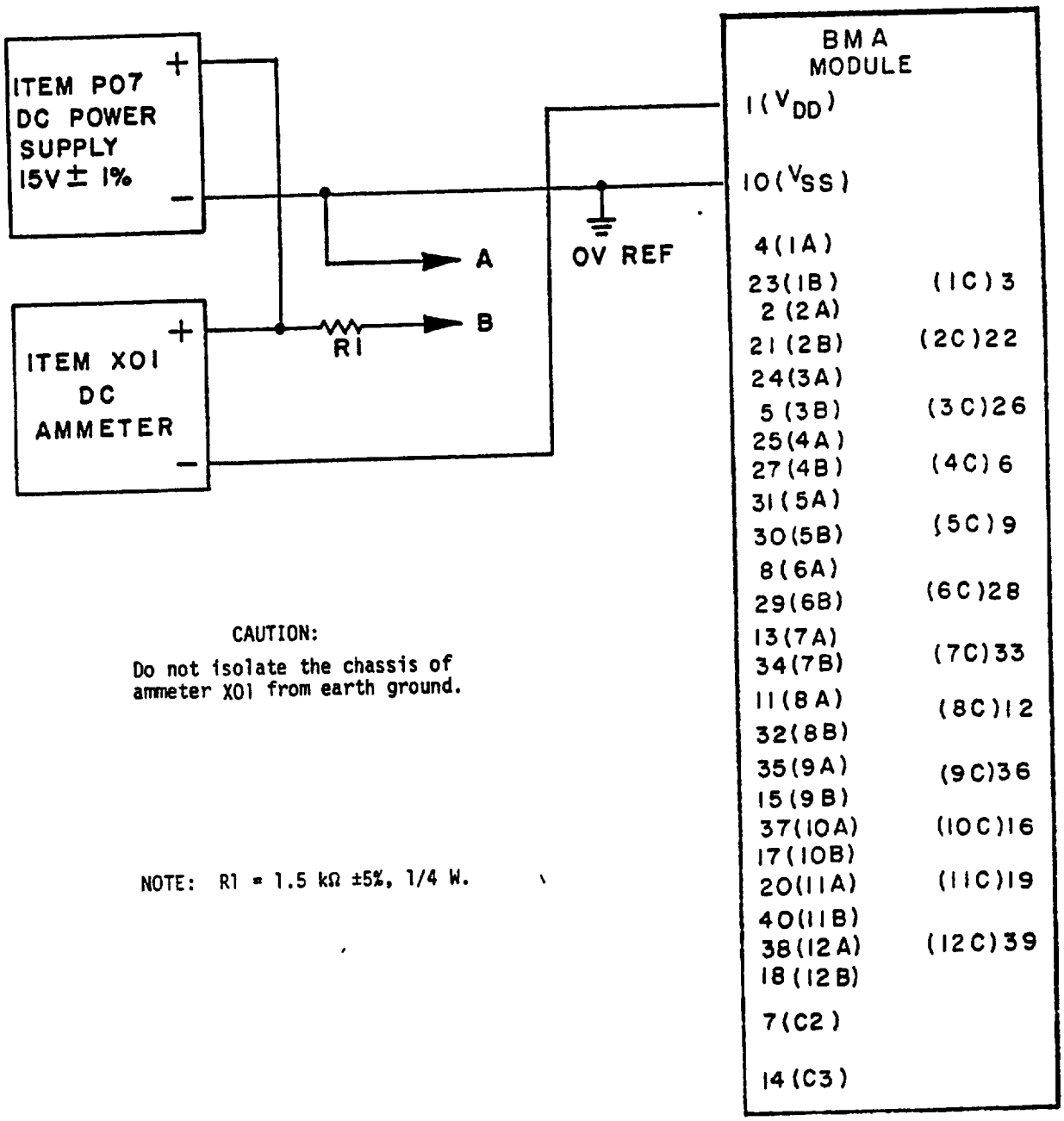
FIGURE 12. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	Open	4,23,2,21,24, 5,25,27,31,30, 8,29,13,34,11, 32,35,15,37,17, 20,40,38,18	0	6	µA	13
M2	Power supply current, high level	4,23,2,21,24,5, 25,27,31,30,8, 29,13,34,11,32, 35,15,37,17,20, 40,38,18	Open	0	6	µA	13

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1A to 1C Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	20	200	ns	14,15
N2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
N3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
N4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
N5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
N6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
N7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
N8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
N9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
N10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
N11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,17,38,18	20	19				
N12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:05Z
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TABLE XV. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N13	1A to 1C Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	68	680	ns	14,15
N14	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
N15	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
N16	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
N17	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
N18	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
N19	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
N20	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
N21	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
N22	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
N23	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
N24	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVI. Transition time (t_{THL}) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P1	1A to 1C	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	20	200	ns	14,15
P2	2A to 2C		4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
P3	3A to 3C		4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
P4	4A to 4C		4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
P5	5A to 5C		4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
P6	6A to 6C		4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
P7	7A to 7C		4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
P8	8A to 8C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
P9	9A to 9C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
P10	10A to 10C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
P11	11A to 11C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
P12	12A to 12C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVI. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P13	1A to 1C	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	68	680	ns	14,15
P14	2A to 2C		4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
P15	3A to 3C		4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
P16	4A to 4C		4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
P17	5A to 5C		4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
P18	6A to 6C		4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
P19	7A to 7C		4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
P20	8A to 8C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
P21	9A to 9C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
P22	10A to 10C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
P23	11A to 11C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
P24	12A to 12C		4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1A to 1C Propagation delay time, (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	25	250	ns	14,15
Q2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Q3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Q4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Q5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Q6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Q7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Q8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Q9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Q10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Q11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Q12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
013	1B to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	25	250	ns	14,15
014	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
015	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
016	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
017	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
018	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
019	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
020	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
021	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
022	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
023	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
024	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q25	1A to 1C Propagation delay time, (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	45	450	ns	14,15
Q26	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Q27	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Q28	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Q29	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Q30	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Q31	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Q32	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 13,34,32,35,15,37,17,20,40,38,18	11	12				
Q33	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Q34	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Q35	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Q36	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q37	1B to 1C Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	45	450	ns	14,15
Q38	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Q39	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Q40	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Q41	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Q42	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Q43	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
Q44	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Q45	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Q46	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Q47	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
Q48	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XVIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1A to 1C Propagation delay time, (t_{PHL}, C_L = 50 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	25	250	ns	14,15
R2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
R3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
R4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
R5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
R6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
R7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
R8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
R9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
R10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
R11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
R12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R13	1B to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	25	250	ns	14,15
R14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
R15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
R16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
R17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
R18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
R19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
R20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,13,35,15,37,17,20,40,38,18	32	12				
R21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
R22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
R23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
R24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R25	1A to 1C Propagation delay time, (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	45	450	ns	14,15
R26	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
R27	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
R28	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
R29	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11 32,35,15,37,17,20,40,38,18	31	9				
R30	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
R31	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
R32	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
R33	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
R34	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
R35	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
R36	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XVIII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R37	1B to 1C Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	45	450	ns	14,15
R38	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,16	21	22				
R39	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
R40	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
R41	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
R42	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
R43	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
R44	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
R45	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
R46	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,36,18	17	16				
R47	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
R48	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XIX. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1A to 1C Transiton time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	10	100	ns	14,15
S2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
S3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
S4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
S5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
S6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
S7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
S8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
S9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
S10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
S11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
S12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XX. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1A to 1C Transiton time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,16	4	3	10	100	ns	14,15
T2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
T3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
T4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
T5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
T6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
T7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
T8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
T9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
T10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
T11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
T12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U1	1A to 1C Propagation delay time, (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	10	100	ns	14,15
U2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
U3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
U4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
U5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
U6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
U7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
U8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
U9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
U10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
U11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
U12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXI. Transition time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U13	1B to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	10	100	ns	14,15
U14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
U15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
U16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
U17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
U18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
U19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
U20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 13,34,11,35,15,37,17,20,40,38,18	32	12				
U21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
U22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
U23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
U24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XXII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	1A to 1C Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	10	100	ns	14,15
V2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
V3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
V4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
V5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
V6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
V7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
V8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
V9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
V10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
V11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
V12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V13	1B to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	10	100	ns	14,15
V14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
V15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
V16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
V17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
V18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,36,18	29	28				
V19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,36,18	34	33				
V20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
V21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,36,18	15	36				
V22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,36,18	17	16				
V23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
V24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XXIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	1A to 1C Transiton time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
W2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
W3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
W4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
W5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
W6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
W7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
W8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
W9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
W10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
W11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
W12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
X1	1A to 1C Transiton time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
X2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
X3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
X4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
X5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
X6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
X7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
X8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
X9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
X10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
X11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
X12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y1	1A to 1C Propagation delay time, (t_{PLH}), C_L = 50 pF, V_{CC} = 15 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
Y2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Y3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Y4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Y5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Y6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Y7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Y8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Y9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Y10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Y11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Y12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y13	1B to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	8	80	ns	14,15
Y14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Y15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Y16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Y17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Y18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Y19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
Y20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Y21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Y22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Y23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,38,18	40	19				
Y24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

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TABLE XXVI. Propagation delay time (t_{PHL}) test.

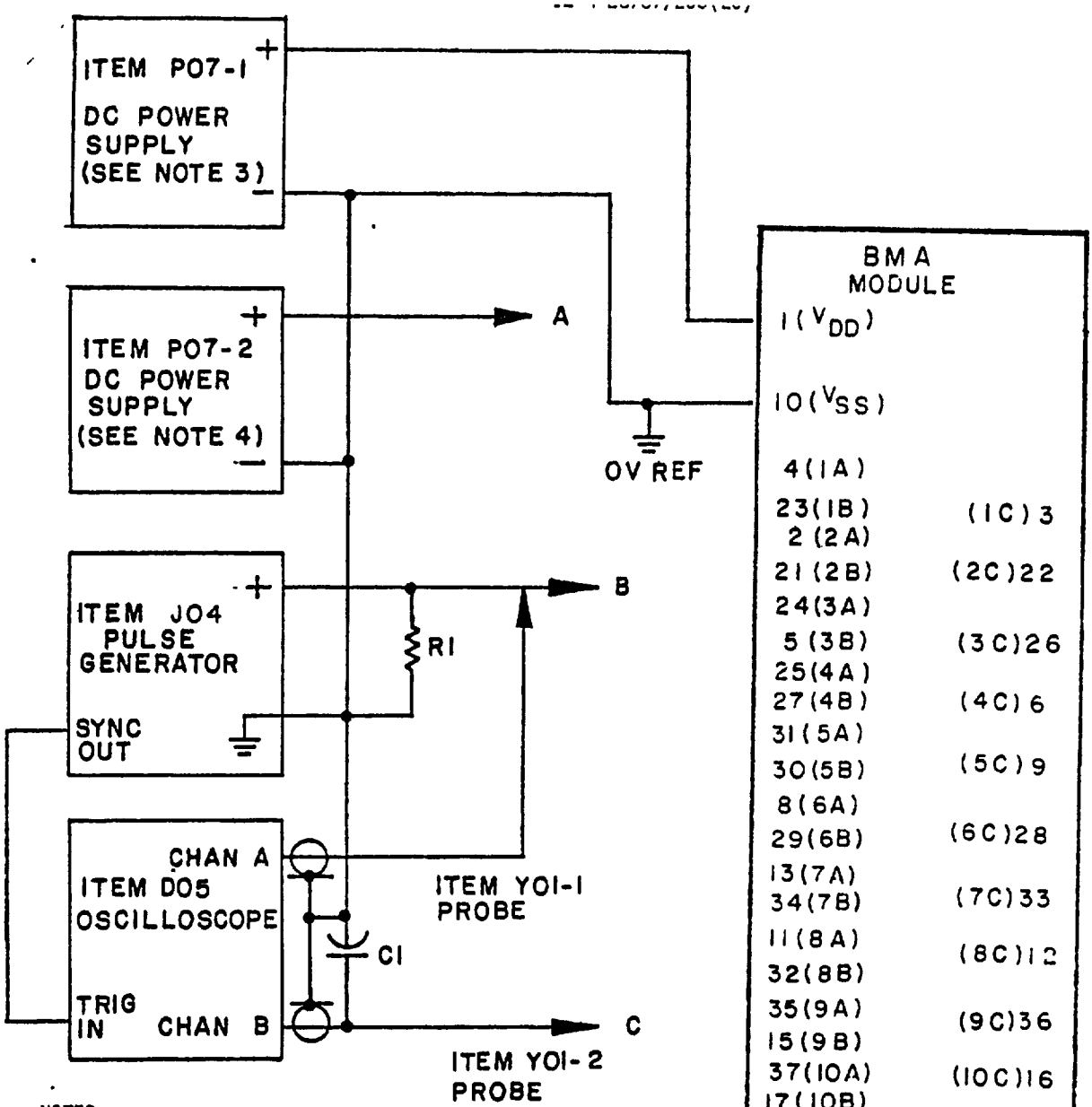
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z1	1A to 1C Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	4	3	8	80	ns	14,15
Z2	2A to 2C	4,23,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	2	22				
Z3	3A to 3C	4,23,21,2,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	24	26				
Z4	4A to 4C	4,23,21,2,5,24,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	25	6				
Z5	5A to 5C	4,23,21,2,5,24,27,25,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	31	9				
Z6	6A to 6C	4,23,21,2,5,24,27,25,30,31,29,13, 34,11,32,35,15,37,17,20,40,38,18	8	28				
Z7	7A to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 34,11,32,35,15,37,17,20,40,38,18	13	33				
Z8	8A to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,35,15,37,17,20,40,38,18	11	12				
Z9	9A to 9C	4,23,21,2,5,24,27,25,30,31,29,6, 34,13,32,11,15,37,17,20,40,38,18	35	36				
Z10	10A to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,20,40,38,18	37	16				
Z11	11A to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,38,18	20	19				
Z12	12A to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,18	38	39				

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TABLE XXVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z13	1B to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = -15 V	4,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	23	3	8	80	ns	14,15
Z14	2B to 2C	4,23,2,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	21	22				
Z15	3B to 3C	4,23,21,2,24,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	5	26				
Z16	4B to 4C	4,23,21,2,5,24,25,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	27	6				
Z17	5B to 5C	4,23,21,2,5,24,27,25,31,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	30	9				
Z18	6B to 6C	4,23,21,2,5,24,27,25,30,31,8,13, 34,11,32,35,15,37,17,20,40,38,18	29	28				
Z19	7B to 7C	4,23,21,2,5,24,27,25,30,31,29,8, 13,11,32,35,15,37,17,20,40,38,18	34	33				
Z20	8B to 8C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,11,35,15,37,17,20,40,38,18	32	12				
Z21	9B to 9C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,35,37,17,20,40,38,18	15	36				
Z22	10B to 10C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,37,20,40,38,18	17	16				
Z23	11B to 11C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,20,38,18	40	19				
Z24	12B to 12C	4,23,21,2,5,24,27,25,30,31,29,8, 34,13,32,11,15,35,17,37,40,20,38	18	39				

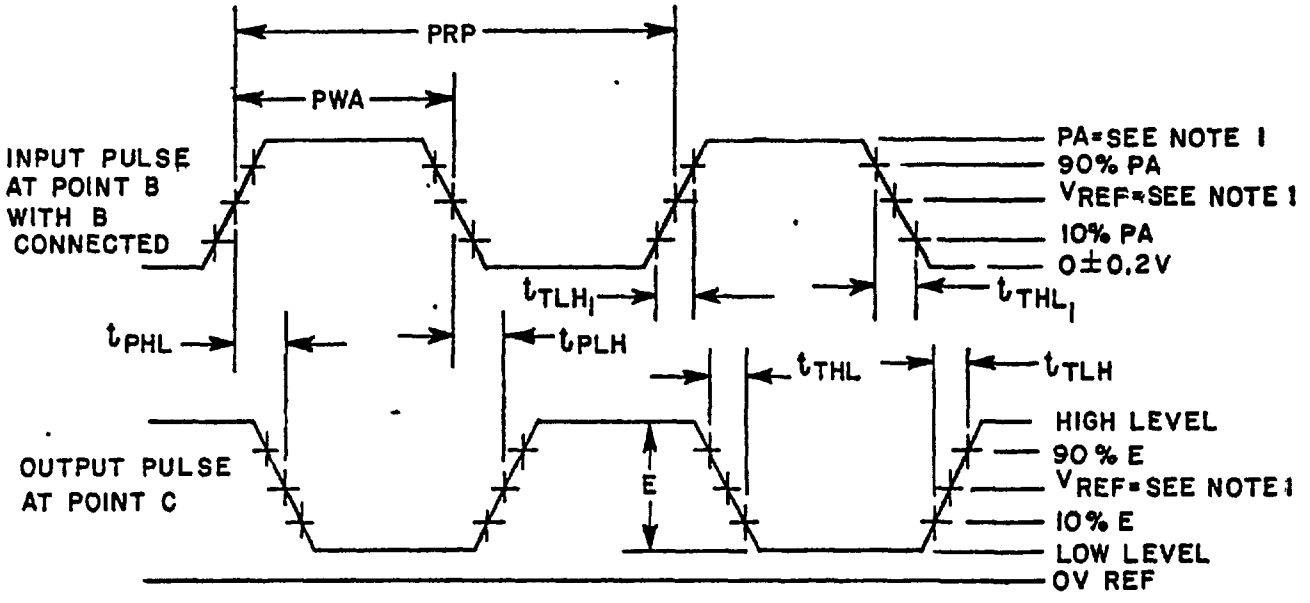
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W (5 V), 3 W (10 V) and 5 W (15 V DC power supply).
2. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N12, P1-P12, Q1-Q24, R1-R24, S1-S12, T1-T12, U1-U24, V1-V24, W1-W12, X1-X12, Y1-Y24 and Z1-Z24.
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions N13-N24, P13-P24, Q25-Q48 and R25-R48.
3. Set item PO7-1 as follows:
   5 V ±1% for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48.
   10 V ±1% for test conditions S1-S12, T1-T12, U1-U24 and V1-V24.
   15 V ±1% for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24.
4. Set item PO7-2 as follows:
   4.5 V ±1% for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48.
   9 V ±1% for test conditions S1-S12, T1-T12, U1-U24 and V1-V24.
   14 V ±1% for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24.

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV through XXVI.

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NOTES:

1. Input pulse parameters:
   PA = 5.0 ±0.1 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48.
   = 10.0 ±0.1 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24.
   = 15.0 ±0.1 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24.
   V_{REF} = 2.5 ±0.1 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48.
   = 5.0 ±0.1 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24.
   = 7.5 ±0.1 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PRP = 5.0 ±0.5 µs.
   PWA = 2.5 ±0.2 µs for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48.
   = 1.0 ±0.1 µs for test conditions S1-S12, T1-T12, U1-U24, V1-V24, W1-W12, X1-X12, Y1-Y24 and Z1-Z24.
2. Before making the measurements, verify that the output pulse high level is ≥ 4.6 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48; ≥ 9.5 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24 and 13.5 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24 and the low level is ≤ 0.4 V for test conditions N1-N24, P1-P24, Q1-Q48 and R1-R48; ≤ 0.5 V for test conditions S1-S12, T1-T12, U1-U24 and V1-V24; ≤ 1.5 V for test conditions W1-W12, X1-X12, Y1-Y24 and Z1-Z24.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV through XXVI.

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TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	0.007	0.013	µF	16

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TABLE XXVIII. C2 and C3 capacitance test.

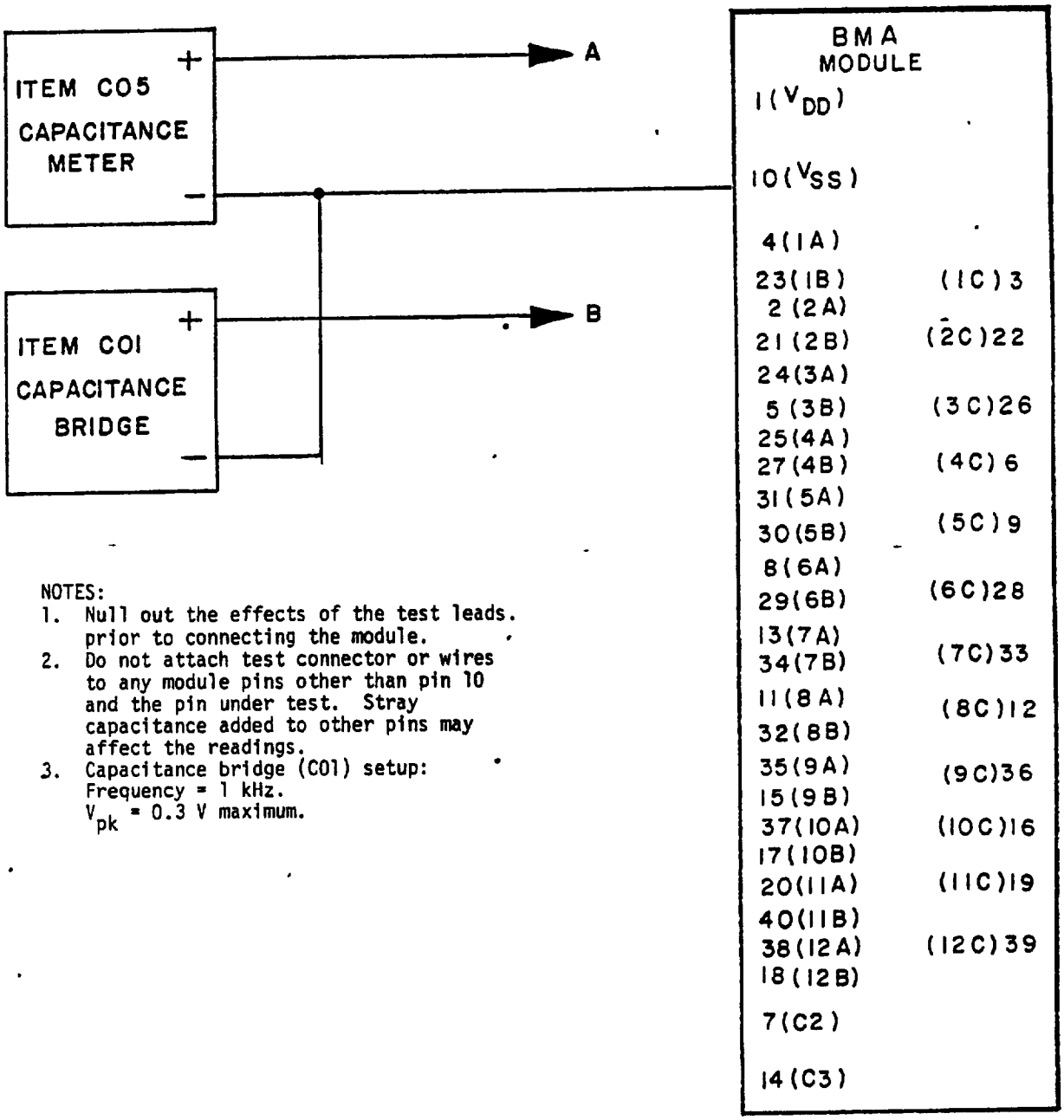
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	C2 capacitance	7	Open	176	264	pF	16
BB2	C3 capacitance	14	Open	176	264	pF	16

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TABLE XXIX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	1A Input capacitance	4	Open	2	15	pF	16
CC2	1B	23					
CC3	2A	2					
CC4	2B	21					
CC5	3A	24					
CC6	3B	5					
CC7	4A	25					
CC8	4B	27					
CC9	5A	31					
CC10	5B	30					
CC11	6A	8					
CC12	6B	29					
CC13	7A	13					
CC14	7B	34					
CC15	8A	11					
CC16	8B	32					
CC17	9A	35					
CC18	9B	15					
CC19	10A	37					
CC20	10B	17					
CC21	11A	20					
CC22	11B	40					
CC23	12A	38					
CC24	12B	18					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (COI) setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 16. Power supply filter, C2, C3 and input capacitance test circuit for tables XXVII, XXVIII and XXIX.

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